

ORIGINAL ARTICLE



Friend of GATA2 Variant Ser657Gly Is Associated With Coronary Microvascular Disease

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BACKGROUND: The coronary microvasculature is crucial for proper cardiac function, and coronary microvascular disease (CMVD) has emerged as an underdiagnosed and undertreated cause of ischemic heart disease. FOG2 (friend of GATA 2) is a transcriptional co-regulator crucial for coronary development and the maintenance of the coronary microvasculature in adult mice. Little is known about the role of FOG2 in humans or its role in CMVD. Here, we report a genotype-first approach to determine the role of FOG2 in human CMVD.

METHODS: We performed phenome-wide association studies and deep cardiac phenotyping through the Electronic Health Record in individuals with FOG2 coding variants. We interrogated MagNET heart tissue data to identify genes and pathways associated with rs28374544. We then overexpressed FOG2S657G in a cardiomyocyte cell line and assessed the effects on cardiac metabolism and paracrine angiogenic signaling.

RESULTS: We identified an association between rs28374544 (A1969G, p.S657G) and CMVD. Using phenome-wide association studies and deep cardiac phenotyping through the Electronic Health Record in individuals with FOG2 coding variants, we identified an association between rs28374544 (A1969G, p.S657G) and CMVD. Individuals carrying the S657G variant, almost all of African ancestry, had increased chest pain, a smaller burden of obstructive coronary artery disease, and altered coronary blood flow. Differential gene and pathway analysis using several genomic data sets showed that carriers of S657G have increased expression of genes involved in angiogenesis, glycolysis, and the HIF (hypoxia-inducible factor) pathway. In vitro functional studies show that compared with the FOG2 wild-type protein, the FOG2^{S657G} variant protein promotes angiogenic gene expression and angiogenesis while decreasing oxygen consumption rate.

CONCLUSIONS: A common functional coding variant in FOG2, S657G, is associated with CMVD in humans. Altered angiogenic gene expression, regulated in part by FOG2, may contribute to CMVD.

Key Words: angiogenesis ■ chest pain ■ gene expression ■ glycolysis ■ oxygen consumption

The coronary microvasculature regulates blood flow to meet cardiac metabolic demand and plays an important role in cardiac health and disease.^{1,2} Coronary microvascular disease (CMVD), defined as disease of the coronary prearterioles, arterioles, and capillaries, can cause angina and myocardial infarction.^{1,2} It also portends a worse prognosis when present with coronary artery disease (CAD)³ and has been implicated in

the pathogenesis of heart failure with preserved ejection fraction.^{4–6} Estimates suggest CMVD may account for 30% to 50% of ischemic heart disease.^{7–9} Yet despite its clinical significance, little is known about the molecular pathways involved in CMVD.

FOG2 (friend of GATA 2, also called zinc-finger protein motif 2 or *ZFPM2*) is a transcriptional coregulator that is crucial for coronary development.¹⁰ Cardiomyocyte

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Nonstandard abbreviations and acronyms

CAD	coronary artery disease
CMVD	coronary microvascular disease
FOG2	friend of GATA 2
HIF	hypoxia-inducible factor
MAF	minor allele frequency
MBF	myocardial blood flow
MTORC1	mammalian target of rapamycin complex 1
PET	positron emission tomography
PheWAS	Phenome-Wide Association Study
PMBB	Penn Medicine Biobank
TFC	TIMI frame count

FOG2 is required for the maintenance of coronary microvasculature in adult mice.¹¹ In particular, in adult mice, FOG2 works in concert with cardiomyocyte GATA4 to promote expression of angiogenic genes (*Vefga*, *Fgfs*) and inhibit inhibitors of angiogenesis (*Timps*)¹¹ and paracrine regulation of the coronary microvasculature. Some studies have linked FOG2 to congenital heart disease,¹² however, little is known about the role of FOG2 on the coronary microvasculature in humans. Here, we aim to determine whether FOG2 regulates the coronary microvasculature in humans and to test the hypothesis that dysregulation of FOG2 may contribute to CMVD.

Using a genotype-first approach focused on coding variants in FOG2 in participants in the Penn Medicine Biobank (PMBB), we found that FOG2^{S657G} (rs28374544) was significantly associated with CMVD. We then used RNAseq data sets from human heart tissue and iPSC lines to show that this variant is gain-of-function for angiogenic gene expression and identified potential pathways differentially regulated by the variant. Finally, we used in vitro studies to support a role for the variant in the regulation of angiogenesis and cardiomyocyte metabolism.

METHODS

Study participants

The PMBB included more than 90 000 participants from the University of Pennsylvania Health System.¹³ This study included 11 017 individuals who had undergone genotyping. Appropriate consent was obtained from each participant regarding storage of biological specimens, genetic sequencing, and access to all available electronic health record data. The study was approved by the institutional review board of the University of Pennsylvania and complied with the principles set out in the Declaration of Helsinki. All individuals included consented to participation in PMBB. Data are available through Gene Expression Omnibus (GSE309526), and the full analytic code used to generate figures is available at <https://github.com/Setia-Verma-Lab/>

Circulation_Genomic_and_Precision-Medicine_Code. Other data that support the findings of this study are available from the corresponding author on reasonable request.

Phenome-Wide Association Study

Phenome-wide association tests were run using the PheWAS (Phenome-Wide Association Study) package in R.¹⁴ Briefly, the PheWAS package uses logistic regression to calculate associations between phenotypes (PheCodes) and a variant of interest.¹⁵ The covariates age, age-squared, sex, and the first 10 PCs from genotypes are adjusted for in the model. We selected phenotypes that have 5% prevalence, which corresponded to 20 cases per PheCode. Results were plotted using the ggplot2 package in R.

Cardiac Phenotyping

Discrete data from coronary angiograms and echocardiograms were obtained for patients who had undergone testing as part of routine clinical care. If more than 1 study was available, then the most complete and recent study was used. *International Classification of Diseases, Ninth or Tenth* codes for chest pain and myocardial infarction were used to identify patients presenting with these clinical syndromes. Perfusion positron emission tomography (PET) stress testing was performed as part of routine medical care as previously described.¹⁶ For deeper phenotyping of matched cohorts, coronary angiogram data were manually abstracted by a blinded operator. The degree of coronary stenosis was assessed in all named coronary arteries and first-order branches. Obstructive disease was defined as $\geq 50\%$ stenosis in the left main coronary artery or $\geq 70\%$ in a named coronary or first-order branch. The Thrombolysis in Myocardial Infarction frame count was performed by a blinded operator as previously described.³ The corrected frame count for all nonobstructed vessels was averaged to represent the frame count for a given patient.

Gene expression and Gene Set Enrichment Analysis

Gene expression data were obtained from the Gene Expression Omnibus Repository (GSE57344) and Database for Genotypes and Phenotypes (phs001341.v1.p1).^{17,18} Partek Flow software (version 10.0.23.0414) was used for differential gene expression analysis (DESeq2), hierarchical clustering and heat map generation, and Gene Set Enrichment Analysis against the hallmark pathway set from the Molecular Signatures Database using the DESeq2 statistic as a ranking metric.¹⁹ Gene Set Enrichment Analysis calculates an Enrichment Score based on how overrepresented a list of ranked genes.¹⁹

Protein Measurements

Plasma was extracted and stored at -80°C before shipping on dry ice to SomaLogic Inc (Boulder, CO) for measurement of the relative concentration of 1305 proteins using the multiplexed, aptamer-based SOMAscan assay. See [Supplemental Methods](#).

In Vitro Assays

Over-expression vectors for FOG2 and empty vector control were purchased (NM_012082, Origene), and Quick Change Lightning site-directed mutagenesis (Agilent, Santa Clara,

CA) was used to generate FOG2 S67G vector. Plasmids were transfected into AC16 cardiomyocytes (SCC109, Sigma Millipore, Burlington, MA) using Lipofectamine 3000 reagent (Invitrogen, Waltham, MA). Gene expression was assayed using quantitative reverse transcription polymerase chain reaction (Applied Biosystems, Thermo Fisher). To test the paracrine effects of cardiomyocyte FOG2 on vascular cells, conditioned media from transfected AC16 cardiomyocytes were used in a tube formation assay. Briefly, HUVECs were seeded onto growth-factor-restricted Matrigel, cultured with conditioned media, and imaged at 4 hours. Oxygen Consumption Rate was measured using a Seahorse Bioanalyzer XF and Mito Stress Test reagents (Agilent, Santa Clara, CA). CRISPR-Cas9 was used to create an FOG2-deficient AC16 cardiomyocyte cell line. Cleavage Under Target and Release Under Nucleosome (CUT&RUN) kit (CST no. 86652) was performed using HIF1a (hypoxia-inducible factor; CST no. 36169S and Abcam no. 2185) or control IgG antibodies (CST no. 66362). Sequencing was performed on NovaSeq, and data were analyzed using Partek Flow software (version 10.0.23.0414). See [Supplemental Methods](#) for additional details.

RESULTS

PheWAS Identifies Association Between FOG2^{S657G} and Clinical Diagnosis of Atherosclerosis

To evaluate for an association between FOG2 and CMVD, we interrogated a cohort of 11 451 participants in the PMBB who had undergone whole-exome sequencing. There were 39 nonsynonymous variants in the coding sequence or 3' untranslated region of FOG2 that were adequately represented in PMBB with at least 20 carriers of the variant ([Table S1](#)). PheWAS were performed for each variant. After correcting for multiple testing, we identified 1 variant, FOG2^{S657G}, that was associated with PheCode for Atherosclerosis ($P < 0.0001$; [Figure 1A](#) and [1B](#)). There were 44 individuals homozygous for the minor allele (GG) and 556 heterozygous for the minor allele (AG), which represents an overall minor allele frequency (MAF) of 0.03 in PMBB. This is comparable to other large heterogeneous databases (GMAF, EXAC); this variant is much more common in populations with African ancestry ([Table S2](#)). Indeed, 41/44 participants who were homozygous for the minor allele self-identified as African-American. GG individuals were younger, more likely to be female, and had increased rates of hypertension and diabetes ([Table 1](#)). We further examined allele frequencies of this variant in 180 whole genome sequences from ethnically diverse African populations,²⁰ as well as in 929 whole genomes from the Human Genome Diversity Project.²¹ We find that globally, the minor allele at rs28374544 (G) is only present at appreciable frequency within Africa ([Figure 1C](#)). Specifically, the variant is at high frequency in some populations from West Africa (Mandenka MAF=0.32; Yoruba MAF=0.25; Tikari MAF=0.23), as well as in the Khoesan

hunter-gatherers from Botswana (IXoo MAF=0.47; Ju|'Hoansi MAF=0.37) and the Herero from Botswana (Herero MAF=0.37). This allele is absent or at low frequency in other African regions ([Figure S1](#)).

The FOG2 protein sequence is highly conserved among species. The Ser657Gly substitution is predicted to introduce a small nonpolar residue into an otherwise hydrophilic ring, potentially altering local protein structure. Using PEP-FOLD3,²² we modeled 18 peptides flanking the variant and found that a glycine substitution introduces additional mobility into a loop domain between zinc fingers ([Figure 1D](#)).

Prior studies have reported an association between rs28374544 and complex congenital heart disease.¹² We used diagnosis codes and natural language processing from coronary angiography and echocardiography reports to determine whether participants who were heterozygous or homozygous for the minor allele had a history of complex congenital heart disease. We found that 1 participant heterozygous for the variant and 2 of the 44 participants homozygous for the variant had a history of Tetralogy of Fallot or Transposition of the Great Arteries and had undergone surgical repair in childhood. This further supports an association between congenital heart disease and rs28374544, with 4.5% of GG having congenital heart disease ($\chi^2=31.2$, $P < 0.0001$; [Table S3](#)).

Clinical Phenotypes of Participants With FOG2 S657G

To gain insight into the association between genotype and the PheWAS association with atherosclerosis, we used the electronic health record to determine clinical cardiac phenotypes. First, for participants with prior clinical echocardiograms, we examined discrete echocardiographic parameters and found no association between genotype and left ventricular size, wall thickness, or ejection fraction ([Figure 2A](#) and [2B](#)). Similarly, there was no association between discrete hemodynamic parameters obtained on left heart catheterization, including aortic blood pressure, heart rate, or left ventricular filling pressures ([Figure 2C](#)). Despite this, participants with GG genotype were more likely to have inpatient admissions with primary diagnosis of chest pain (AA, 717/10138; AG, 92/464; GG, 10/34; χ^2 $P < 0.0001$; [Figure 2D](#)) or myocardial infarction (AA, 705/10150; AG, 39/517; GG, 6/38; χ^2 $P < 0.0001$; [Figure 2E](#)).

Cardiac stress testing with rubidium-82 perfusion PET imaging is routinely performed at the Hospital of the University of Pennsylvania for clinical indications, including the evaluation of chest pain. In addition to myocardial perfusion imaging, Rb-82 perfusion PET imaging provides quantitative myocardial blood flow (MBF) measurements at rest and under hyperemic, or cardiac stress, conditions. The ratio of stress MBF to rest MBF is the MBF reserve, which, in the absence of coronary

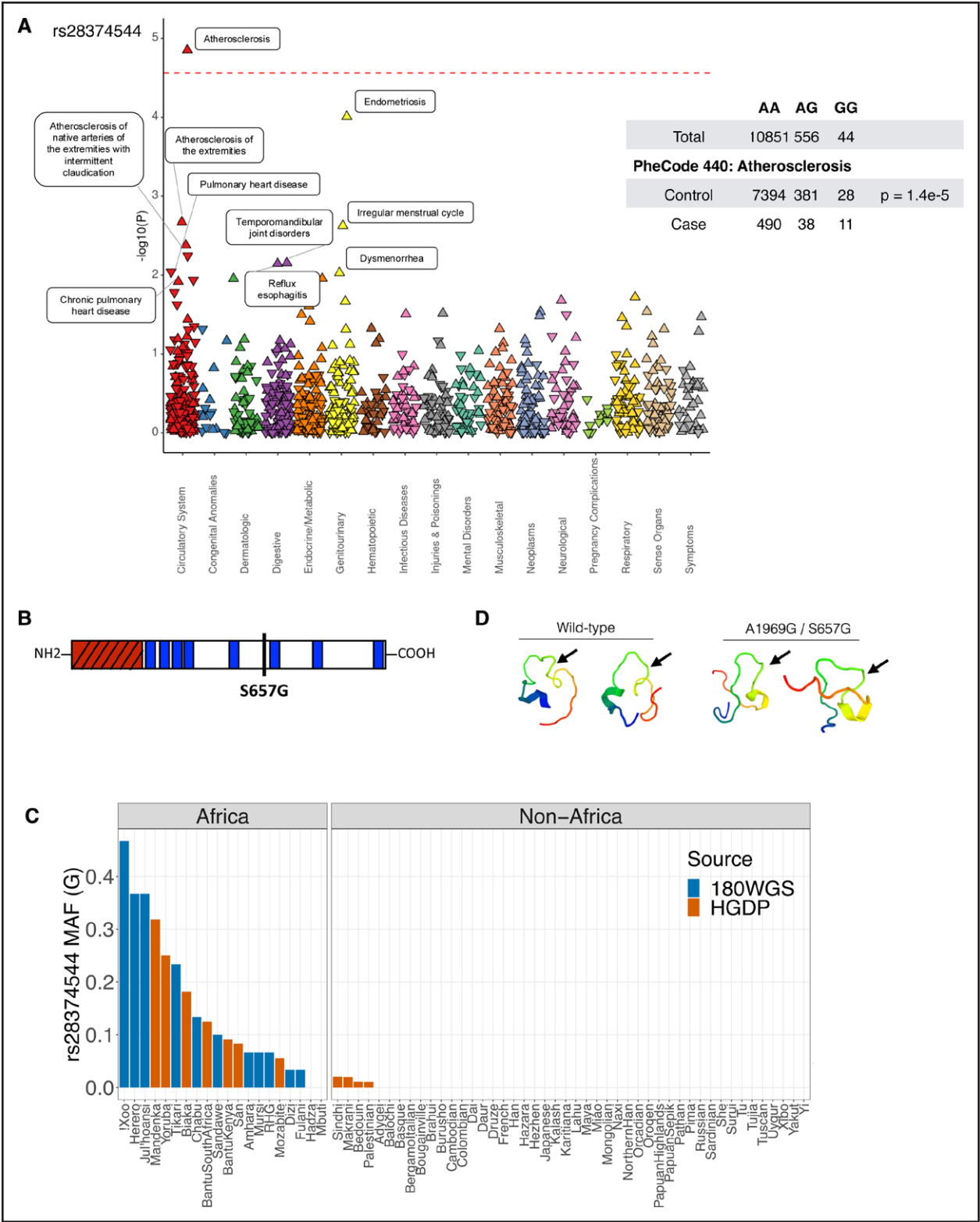


Figure 1. PheWAS (Phenome-Wide Association Study) identified an association between rs28374544 and a coronary vascular phenotype.
A, PheWAS analysis identified an association between rs28374544 and the Phecode for atherosclerosis, red line=Bonferroni threshold for statistical significance. Summary table of the cases and controls of PheCode 440 in cases and controls. **B**, Schematic of FOG2 (friend of GATA 2) protein highlighting the location of the coding variant relative to zinc finger domains (blue) and repressor domain (red, striped). **C**, Minor allele frequency (MAF) for the variant is high in African populations and near absent in non-African populations (whole genome sequencing [WGS] from [13], Human Genome Diversity Project [HGDP] from [14]). **D**, PEP-FOLD3 modeling showing local ring structure, which is disrupted with S657G substitution.

Table 1. PMBB Demographics by Genotype

	Overall (N=11017)	AA (N=10436)	AG (N=537)	GG (N=44)	P value*
Age	70 (61–79)	71 (62–79)	64 (55–72)	67 (57–77)	<0.001
Sex					<0.001
Female	4523 (41%)	4231 (41%)	265 (49%)	27 (61%)	
Male	6494 (59%)	6205 (59%)	272 (51%)	17 (39%)	
Race					
American Indian and Alaska Native	1 (<0.1%)	1 (<0.1%)	0 (0%)	0 (0%)	
Asian	139 (1.3%)	139 (1.3%)	0 (0%)	0 (0%)	
Black	2098 (19%)	1558 (15%)	498 (93%)	42 (95%)	
Hawaii Pacific Islander	5 (<0.1%)	5 (<0.1%)	0 (0%)	0 (0%)	
Other	364 (3.3%)	356 (3.4%)	8 (1.5%)	0 (0%)	
Unknown	404 (3.7%)	395 (3.8%)	9 (1.7%)	0 (0%)	
White	8006 (73%)	7982 (76%)	22 (4.1%)	2 (4.5%)	
Coronary atherosclerosis	4360 (48%)	4174 (48%)	167 (40%)	19 (49%)	0.008
Diabetes	3174 (34%)	2915 (32%)	236 (52%)	23 (56%)	<0.001
Heart Failure	3752 (42%)	3542 (42%)	197 (43%)	13 (37%)	0.7
Hyperlipidemia	5930 (64%)	5606 (64%)	298 (64%)	26 (65%)	>0.9
Hypertension	6442 (69%)	6031 (69%)	378 (79%)	33 (80%)	<0.001
Ischemic heart disease	4883 (51%)	4645 (51%)	214 (46%)	24 (55%)	0.2

Median (IQR); n (%). IQR indicates interquartile range; and PMBB, Penn Medicine Biobank.
*Kruskal-Wallis rank-sum test; Pearson χ^2 test; Fisher exact test.



stenosis, reflects coronary microvascular function. Since self-identified race is known to affect MBF reserve,¹⁶ we examined MBF values for Black participants with AA and AG genotype in the PMBB. We found that individuals heterozygous for the variant had increased $\approx$ 30% increased resting MBF (mean, 1.02 versus 1.36 mL/min per gram [95% CI, 0.11–0.57]; $P<0.005$) and increased stress MBF (mean, 1.85 versus 2.47 mL/min per gram [95% CI, 0.17–1.07]; $P<0.01$), with no change in MBF reserve (Figure 2F). This suggests a genotype-specific effect on MBF.

Cardiac Phenotyping of Matched Cohorts Reveals Association Between FOG2^{S657G} and CMVD

To perform more in-depth cardiac phenotyping, we started with 44 GG participants and identified cohorts of 88 AA and 88 AG individuals that were matched based on age, sex, and self-reported race (Table 2). The groups had a similar distribution of cardiovascular risk factors and differed only in the diagnosis of atherosclerosis, which is consistent with the PheWAS analysis.

To determine whether GG individuals might have increased CAD as the cause of increased chest pain and MI, we assessed the burden of CAD based on manually abstracted coronary angiograms by a blinded reviewer. Obstructive CAD was defined as $\geq$ 70% stenosis in a named coronary artery or first-order branch or $>$ 50% stenosis in the left main coronary artery. GG

participants had a lower burden of obstructive CAD (Figure 2G). Conversely, to evaluate for altered coronary microvascular flow and function, we assessed Thrombolysis in Myocardial Infarction frame count (TFC), a quantitative measure of the time it takes contrast dye to travel from a proximal landmark to a distal landmark during a coronary angiogram cine that has been previously used as a measure of the coronary microvasculature.^{3,23} Measuring TFC only in nonobstructed vessels, we found that participants with the reference allele had TFC of 14.6 ± 3.88 , whereas those heterozygous or homozygous for the variant had TFC of 16.46 ± 4.17 and 13.47 ± 3.20 , respectively (ANOVA $P=0.039$; Figure 2H). A clinical diagnosis of CMVD is based on (1) the presence of symptoms of cardiac ischemia, (2) without obstructive coronary artery disease or other obvious alternate cardiac cause, and requires (3) objective documentation of cardiac ischemia and, ideally, (4) confirmatory testing showing coronary flow reserve.²⁴ Given the limitations of electronic health record data, we assessed participants for the presence of the first 2 criteria and found that 6.8% of those homozygous for the variant met CMVD criteria, whereas 2.3% and 0% of those heterozygous or without the variant, respectively, met criteria (Figure 2I). Though not all of these individuals have CMVD, these data support the association between the variant and CMVD. Furthermore, these data suggest that the chest pain and MI signals noted at the population scale may be due to CMVD rather than obstructive CAD.

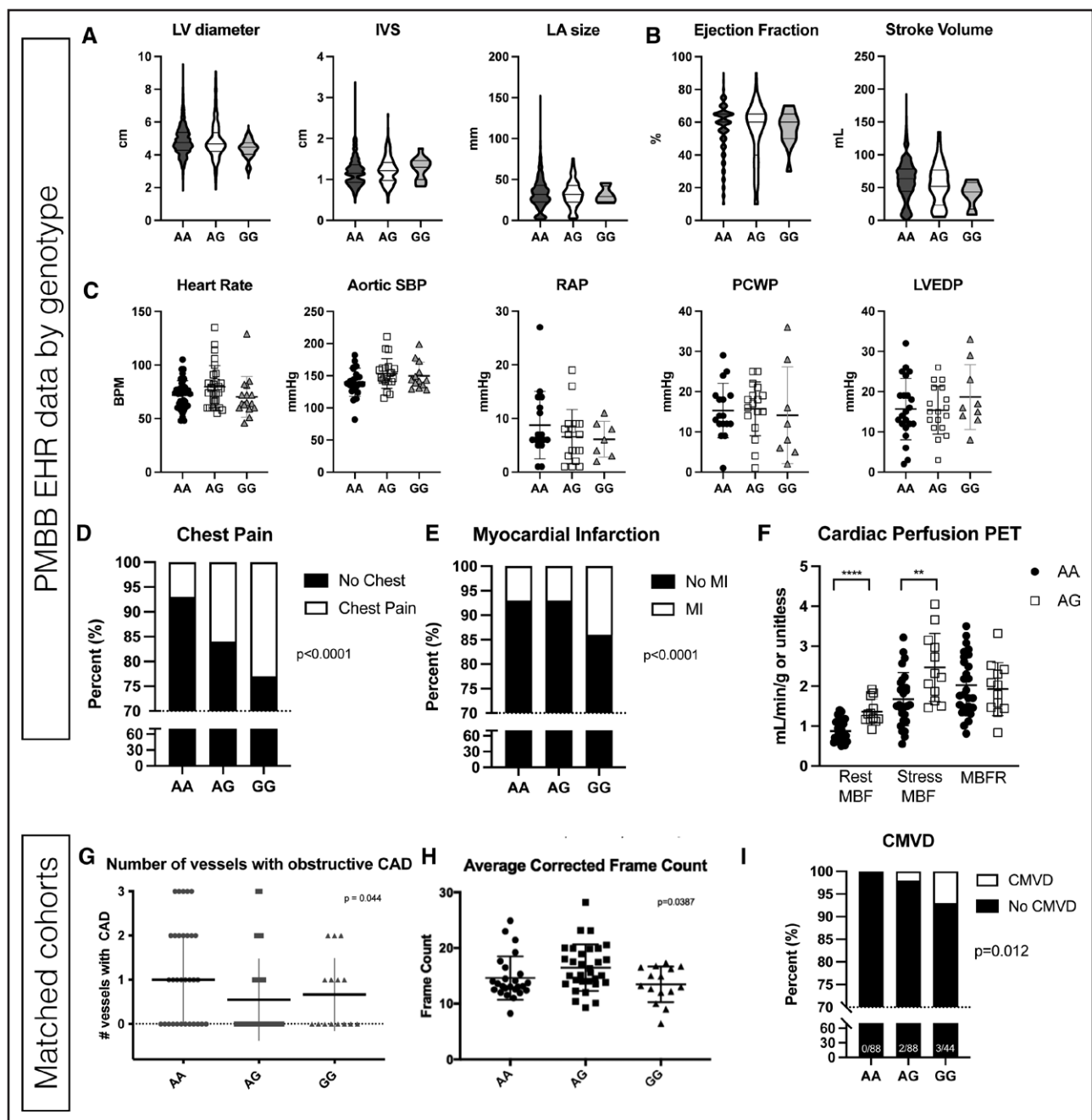


Figure 2. Clinical phenotyping of individuals with and without FOG2 (friend of GATA 2) S657G variant.

A and **B**, Cardiac structure (left ventricular [LV] diameter and interventricular septal [IVS] wall thickness and left atrial [LA] volume) and function (ejection fraction and stroke volume) assessed by echocardiography ($n=4748$, 269, and 29 for AA, AG, GG, respectively). **C**, Invasive hemodynamic parameters assessed during cardiac catheterization, including heart rate, aortic systolic blood pressure (SBP), right atrial pressure (RAP), pulmonary capillary wedge pressure (PCWP), and LV end-diastolic pressure (LVEDP; $n=22$, 18, and 9 for AA, AG, GG, respectively). **D** and **E**, Percent of inpatient primary diagnosis of chest pain and myocardial infarction by genotype. **F**, Perfusion cardiac PET-measured quantitative flow. **G**, Number of vessels with obstructive coronary artery disease (CAD) in named coronary artery or first-order branch by genotype. **H**, Average corrected Thrombolysis in Myocardial Infarction (TIMI) frame count by genotype. **I**, Percent of individuals meeting clinical criteria for coronary microvascular disease (CMVD) by genotype. **A–C**, **G**, **H**, ANOVA analyses; **D**, **E**, χ^2 test; $^*P<0.005$, $^{**}P<0.01$, $^{****}P<0.001$.

Genomic Analysis of FOG2^{S657G} in Human Heart Tissue and Induced Pluripotent Stem Cell Lines

We next sought to assess the effects of FOG2^{S657G} on gene expression in human heart tissue. We used

RNAseq and genotyping data from control hearts from Black participants from the MAGnet study (Figure 3A).¹⁷ We further subset by sex and focused on females since there were more female samples and a majority of our human cohort was female. First, we

Table 2. Cohorts Matched on Age, Sex, and Ancestry

	AA (N=88)	AG (N=88)	GG (N=44)
Age	58.3 (15.0)	58.0 (15.0)	58.0 (15.2)
Female	54 (61.4)	54 (61.4)	27 (61.4)
African-American	82 (93.2)	82 (93.2)	41 (93.2)
Body mass index, kg/m ²	32.7 (8.6)	31.3 (7.2)	32.9 (9.5)
Hypertension	76 (86.4)	71 (80.7)	35 (79.5)
History of myocardial infarction	2 (2.3)	4 (4.5)	1 (2.3)
History of prior CAD	37 (42.0)	38 (43.2)	22 (50)
Diabetes	0 (0)	0 (0)	0 (0)
Chronic kidney disease	8 (9.1)	8 (9.1)	3 (6.8)

Median (IQR); n (%). CAD indicates coronary artery disease; and IQR, interquartile range.

established the effect of FOG2^{S657G} on known FOG2 target genes. Unbiased clustering of genes from Zhou et al¹¹ showed that angiogenic and antiangiogenic gene targets of FOG2 clustered separately, with the exception of *FGF16* and *COL4A3* (Figure 3B). AG and GG genotypes were associated with increased expression of angiogenic genes, including *VEGFA*, *FGF2*, *FGF9*, and decreased expression of inhibitors of angiogenesis such as *TIMP1* and *TIMP2*. These data suggest that FOG2^{S657G} may act as a gain-of-function variant for angiogenic genes. We next performed differential gene expression analysis comparing AA females with AG and GG females. Gene set enrichment analysis of differentially expressed genes identified several pathways as differentially regulated in individuals with rs28374544, including unfolded protein response, hypoxia, and MTORC1 (mammalian target of rapamycin complex) pathways (Figure 3C).

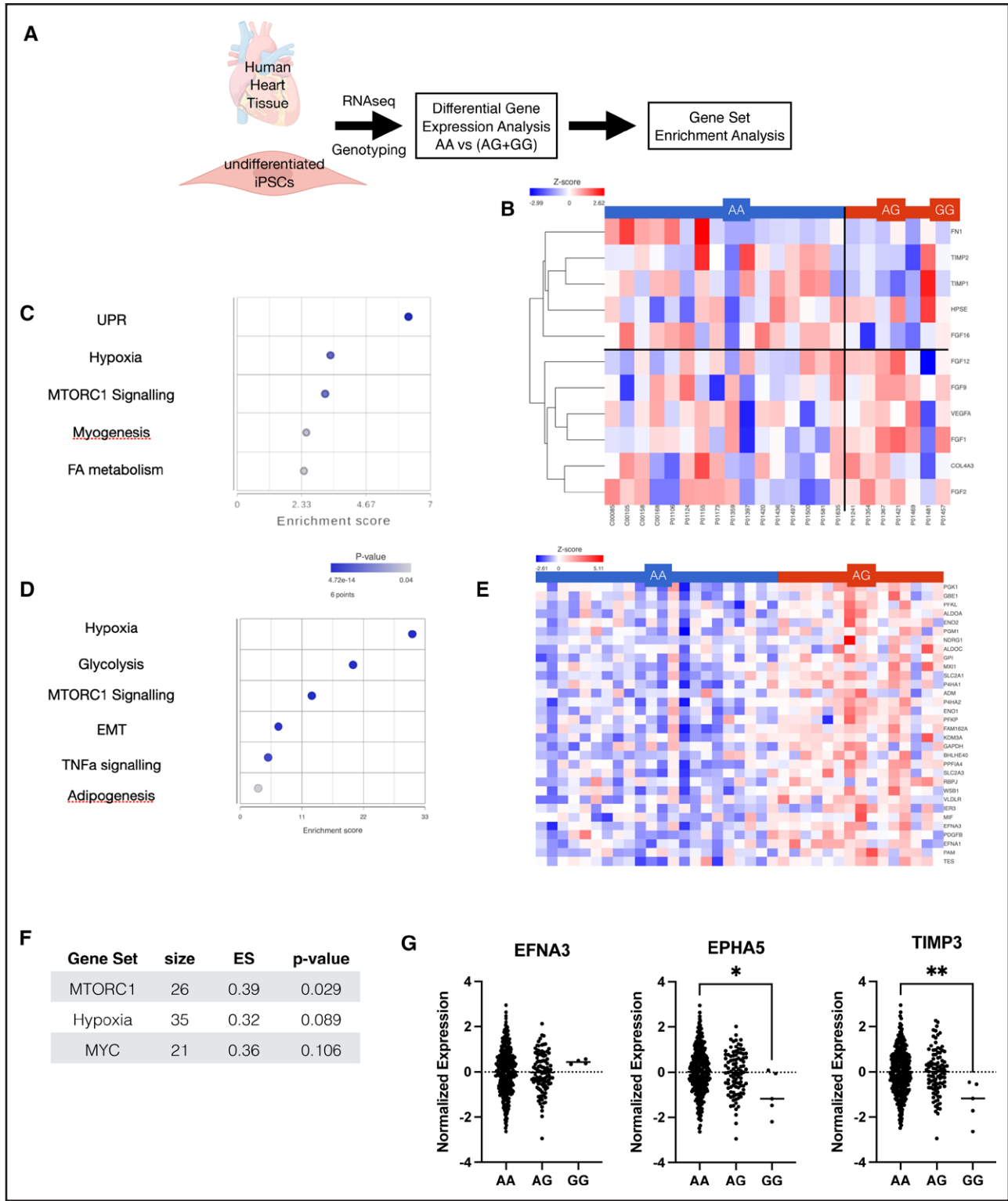
To validate the association between the variant and hypoxia and MTORC1 pathways, and also to determine whether these changes represented fundamental cell-autonomous processes or were the result of interactions of multiple cell types, we assessed the effects of FOG2^{S657G} on gene expression in a cohort of previously reported undifferentiated human iPSC lines.¹⁸ We analyzed the RNAseq data from 25 cell lines isolated from Black female participants: 9 lines from carriers homozygous for the variant and 16 lines from individuals without the variant. Gene-set enrichment analysis identified hypoxia, glycolysis, and MTORC1 pathways as the most significantly enriched gene sets (Figure 3D). Several of the hypoxia-related genes were increased in AG, suggesting that FOG2 may contribute to the regulation of hypoxia-inducible genes. Several angiogenic and hypoxia-inducible gene targets were increased in AG females, including *ADM*, *EFNA1*, *EFNA3*, and *PDGFB*. Several hypoxia-regulated genes, which play a role in metabolism, were also increased in AG females, including *ALDOA*, *ALDOC*, *PFKL*, *PGK1*, and *ENO2*.

Lastly, we examined the levels of serum proteins in 389 PMBB participants of African ancestry who had undergone proteomic profiling with SOMAscan Somalogic and genotyping. The mean age of participants was 61±13.6, and 58% were male. For this exploratory analysis, we compared the top plasma proteins differentially regulated between AA and AG versus GG, and performed gene set enrichment analysis. MTORC1 was the most enriched pathway, and hypoxia was second, though not statistically significant (Figure 3F). *EFNA3*, which was increased in iPSCs, showed a trend towards an increase in GG participants. *EPHA5*, a receptor for *EFNA3*, was downregulated in participants homozygous for the variant. *TIMP3*, an antiangiogenic gene, was decreased in the serum of GG participants (Figure 3G).

FOG2^{S657G} Regulates Hypoxia-Induced Gene Expression

To determine whether the effects of the FOG2^{S657G} variant may be due to the variant alone rather than other variants that may be in linkage disequilibrium, we turned to an in vitro system. FOG2 is more highly expressed in cardiomyocytes, and cardiomyocyte expression is crucial for the paracrine regulation of vascular cells.¹¹ We, therefore, used AC16 cardiomyocytes, an immortalized human left ventricular cardiomyocyte cell line,²⁵ and lipid-based transfection to overexpress control empty vector, FOG2, or FOG2^{S657G}. We saw no significant differences under basal conditions with 21% oxygen. Based on genomic data suggesting FOG2 may be involved in hypoxia-related pathways, we treated cells with low-dose dimethyloxalylglycine (DMOG), a chemical stabilizer of HIF proteins. We then noted an increase in expression of angiogenic genes *VEGFA* and *ADM* in cells treated with FOG2^{S657G} relative to wild-type FOG2. *ANGPT1*, which is known to decrease angiogenesis and promote vessel maturation, was decreased by FOG2^{S657G} (Figure 4A). To assess the paracrine effect of these gene changes on angiogenesis, we used conditioned media from the AC16 cells in a tube formation assay with human umbilical vein endothelial cells. Human umbilical vein endothelial cells treated with conditioned media from FOG2^{S657G}-overexpressing cardiomyocytes formed tube networks with more junctions and segments and longer branches and total length (Figure 4B and 4C), consistent with increased angiogenic potential.

Genomic data suggested that FOG2^{S657G} promoted expression of glycolytic genes. To determine whether FOG2^{S657G} had an effect on oxidative phosphorylation, we measured oxygen consumption rate. FOG2^{S657G}-overexpressing cells had decreased basal respiration. Following FCCP administration to measure maximal oxygen respiration, FOG2^{S657G}-overexpressing cells had a lower oxygen consumption rate.



FOG2 Regulates Expression of Angiogenic HIF1a Targets

Genomic data suggested a link between FOG2^{S657G} and hypoxia-regulated pathways, and in vitro studies suggested FOG2^{S657G} may regulate HIF-induced angiogenesis and oxidative phosphorylation. Though FOG2 is known to interact with GATA4²⁶ and additional transcription factors, including COUP-TFII,²⁷ it is now known whether FOG2 can regulate HIF1a-mediated gene transcription. We used CRISPR-Cas9 gene editing to generate AC16 cardiomyocyte cell lines deficient in FOG2. We targeted exon 8 and established a cell line with reduced expression of both FOG2 and isoform FOG2S (Figure 5A). To determine whether FOG2 affected HIF1a-mediated transcription, we treated cells with DMOG, which stabilizes HIF1a and leads to HIF1a-mediated gene transcription. In particular, known angiogenic targets of HIF1a, *VEGFA*, and *ADM* showed increased transcription over time. However, cells deficient in FOG2 showed a blunted response relative to control (Figure 5B). In contrast to these angiogenic HIF1a targets, transcription of metabolic gene targets *PDK1* and *BNIP3* was similar in control and FOG2 knockdown cell lines (Figure 5C). We next performed cleavage under targets and release using nuclease (CUT&RUN) to profile HIF1a-chromatin interactions in the promoter region of these genes. HIF1a occupancy was noted in several regions of *VEGFA* and *ADM* promoters, consistent with the presence of many HIF-responsive elements. However, we saw a decrease in HIF1a binding in FOG2KD cell lines in specific regions, also marked by HIF-responsive elements (red box, Figure 5D). In contrast, HIF1a occupancy in the promoters of *PDK1* and *BNIP3* was similar in control and FOG2 KD cells. These data suggest that FOG2 may serve as an angiogenic-gene-specific coregulator of HIF1a.

DISCUSSION

We used a genotype-first approach to show an association between FOG2^{S657G}, a variant present in individuals of African ancestry, and clinical findings consistent with CMVD. Using gene expression data from human heart samples, we show that FOG2^{S657G} may serve as a gain-of-function variant for angiogenic gene expression. Unbiased distinct analyses using human heart tissue, undifferentiated iPSCs, and plasma proteomic data suggest that FOG2^{S657G} is associated with gene changes related to hypoxia, MTORC1, and unfolded protein

response pathways. In vitro functional studies support a role for FOG2^{S657G} regulation of hypoxia-induced angiogenic gene expression, tube formation, and cellular metabolism.

The clinical diagnosis of CMVD is challenging, and retrospective analyses using electronic health record data are particularly limited. However, using deep cardiac phenotyping in individuals with the variant, we found a pattern consistent with CMVD. In particular, individuals with FOG2^{S657G} were more likely to present with chest pain or MI, yet they did not have a lower burden of obstructive CAD paired with altered coronary flow by both TFC and perfusion PET. These data suggest that the increased chest pain or MI may have been caused by CMVD rather than obstructive CAD. Of note, both PET and TFC data suggested increased coronary flow in individuals with the FOG2^{S657G} variant. Given the need to tightly match blood supply and demand, increased flow may represent a dysregulation that drives symptoms over time. Similarly, we did not observe a difference in flow reserve by PET. This may be because the study is underpowered or, alternatively, FOG2^{S657G} may lead to structural abnormalities rather than coronary microvascular dysfunction.²⁸ For example, altered angiogenic signaling may lead to a microvascular network with dilated vessels or more collateralization, resulting in increased flow under rest and stress conditions and lower TFC.

Based on 3 distinct analyses, the top differentially regulated pathways associated with FOG2^{S657G} were hypoxia and MTORC1. We have previously shown that FOG2^{S657G} can activate the MTORC1 pathway in hepatocytes,²⁹ a master regulator of multiple pathways including angiogenesis and metabolism. Though the mechanism by which FOG2^{S657G} leads to increased risk of CMVD is unknown, 1 potential explanation may be that increased angiogenic signaling disrupts proper vessel formation or maintenance. For example, Pontes-Quero et al,³⁰ showed that increased angiogenic signals can lead to dysregulation of angiogenesis. Another possibility is that FOG2^{S657G} impairs cardiac metabolism and disrupts the tight coupling of blood flow and cardiomyocyte metabolism. Any mismatch between blood supply and cardiac demand can result in cardiac ischemia and symptoms of CMVD. Both of these mechanisms could contribute to the phenotype. Additional mechanistic studies in preclinical models are needed to untangle the relative contributions of these to the coronary microvasculature and cardiac metabolism.

Figure 3 Continued. and *ENO2*. *EFNA3* was also increased in this gene set (n=22 AA and 15 AG). **F**, Top gene sets identified using a previously reported plasma protein data set also identified MTORC1 with a nominal *P* value of 0.029. ES=GSEA enrichment score, size=number of differentially regulated genes present in gene set. **G**, Normalized protein expression for angiogenic protein *EFNA3* suggests a trend towards an increase. *EPHA5* and *TIMP3* were decreased (n=287 AA, 97 AG, and 5 GG), ANOVA with multiple comparison testing (**P*<0.05, ***P*<0.01). EMT indicates epithelial-to-mesenchymal transition; FA, fatty acid; and UPR, unfolded protein response,

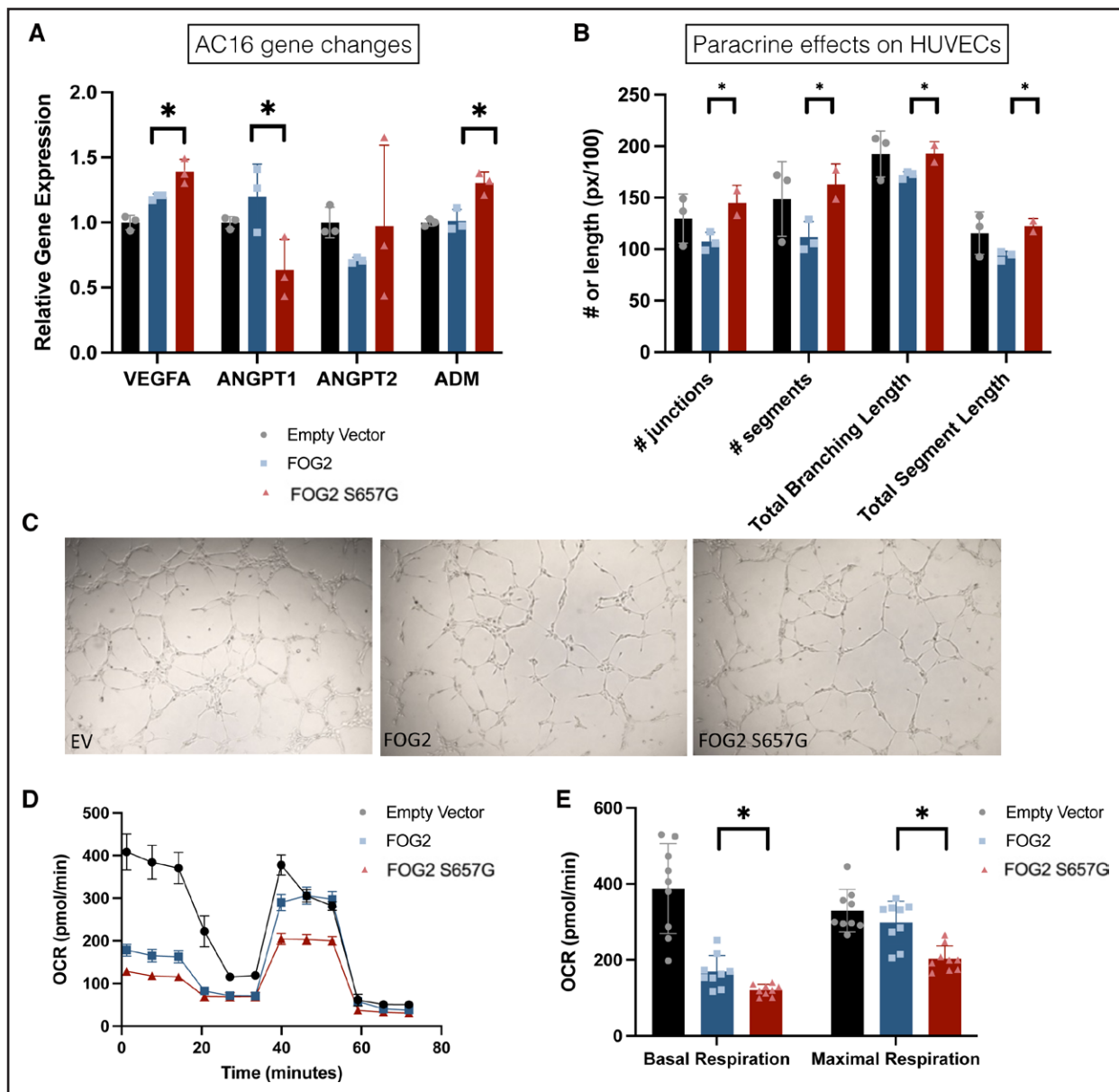


Figure 4. FOG2 (friend of GATA 2) S657G regulates angiogenesis and metabolism in vitro.

A, Angiogenic gene expression in AC16 cardiomyocytes overexpressing FOG2^{S657G} relative to wild-type FOG2. **B** and **C**, Tube formation angiogenesis assay with HUVECs incubated with conditioned media from cells treated with FOG2^{S657G}, wild-type FOG2, or control empty vector. **D** and **E**, Oxygen consumption rate measured during a mito stress test in AC16 cells (n=9 per condition). ANOVA analyses with multiple comparison testing; **P*<0.05).

The mechanism by which FOG2 regulates gene expression is not clear. On one hand, FOG2 is known to interact with GATA4, and the S657G variant has been shown to reduce the ability of FOG2 to repress GATA4-mediated transcription.¹² As such, FOG2 serves as a transcriptional co-repressor, possibly via recruitment of a repressor complex.³¹ Conversely, Zhou et al showed that FOG2 interaction with GATA4 is necessary for regulation of angiogenic FOG2 targets, such that disrupting GATA4-FOG2 interactions decreased target gene expression. Thus, FOG2 regulation must be multifactorial and likely involve interactions with other

cardiac transcription factors. In our studies, we found that many hypoxia-inducible genes are regulated by FOG2. We also show that a FOG2-knockout cell line has a blunted response to hypoxia-mimicking stimulus leading to decreased HIF1a occupancy in target gene promoters. These data suggest FOG2 may serve as a cofactor for hypoxia-inducible factor 1a. Additional studies are needed to understand these interactions.

FOG2^{S657G} is enriched in populations of African ancestry. It is well established that Black individuals suffer from a higher burden of cardiovascular disease and worse cardiovascular outcomes.³² Similarly, Black

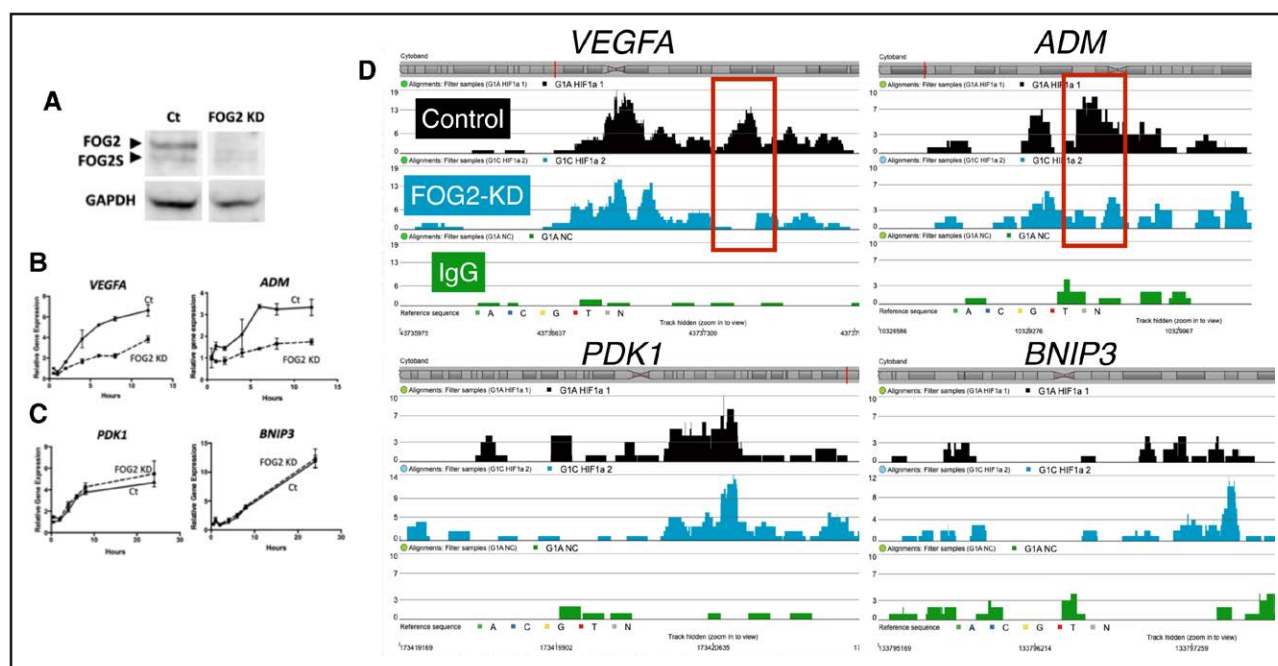


Figure 5. FOG2 (friend of GATA 2) knockdown (KD) blunts hypoxia-induced angiogenic gene expression.

A, Western Blot showing decreased FOG2 in the FOG2-KD cell line, generated using CRISPR-Cas-mediated disruption of exon 8, which is shared by FOG2 and FOG2S. **B**, When cells are exposed to hypoxia, FOG2 KD show decreased expression of angiogenic genes *VEGFA* and *ADM* over time relative to control ($n=3$ /group). **C**, Gene expression of *PDK1* and *BNIP3*, nonangiogenic targets of HIF1 α (hypoxia-inducible factor), is not affected in FOG2-KD ($n=3$ /group). **D**, CUT&RUN with DNA sequencing shows HIF1 α binding to the promoters of *VEGFA*, *ADM*, *PDK1*, and *BNIP3*. FOG2KD decreased HIF1 α occupancy of angiogenic gene promoters relative to control (red boxes). Representative image of $n=3$ /group is displayed.

women with INOCA in the Women's Ischemia Syndrome Evaluation Study had a higher incidence of major adverse cardiovascular events.³³ We have previously reported that Black individuals may have higher coronary flow reserve,¹⁶ though these data are retrospective and do not adequately account for confounders. Others have found that Black participants have reduced coronary flow reserve in the context of left ventricular hypertrophy.³⁴ Additional studies are needed to establish the prevalence of CMVD in distinct populations and how they contribute to a larger burden of adverse cardiovascular outcomes. Although these differences are largely driven by social determinants of health, it is also possible that underlying genetic variation may interact with environmental factors to promote CMVD. Understanding the mechanisms driving these differences may help identify novel solutions and chip away at health disparities.

There are several limitations to this work. It is a single-center study performed at a tertiary care center. Replicating these results in other biobank data sets and populations is crucial. Additionally, though our results suggest a link with human phenotype and functional studies support a role for FOG2^{S657G} in mediating hypoxia-related pathways, it is also possible that FOG2^{S657G} is in linkage disequilibrium with other variants that regulate FOG2 expression or protein function direction or regulate expression of other genes. We also do not directly

show that cardiac expression of FOG2^{S657G} is driving the phenotype seen in participants with the variant. FOG2 is expressed in a few tissues, including the heart, liver, cerebellum, and ovaries, in the genotype-tissue expression project.³⁵ Our in vitro studies outline a potential mechanism by which FOG2 may affect cardiomyocyte gene expression and result in a CMVD-like phenotype. However, in vivo studies are needed to definitively establish the role of FOG2^{S657G} on the coronary microvasculature and fully understand the interplay between cardiac cell types. Lastly, we outline a potential role whereby FOG2^{S657G} may lead to dysregulation of angiogenic factors, which may affect the coronary microvasculature. This could affect coronary microvascular structure and function in adults, but it is also possible that the variant results in altered coronary patterning during development. Additional studies are needed to understand whether FOG2^{S657G} may affect coronary microvascular development.

In summary, we examined FOG2 variants in the PMBB and identified a coding variant in the FOG2 gene, which is predominantly present in individuals of African descent and is associated with a CMVD-like phenotype. These results establish an association between the FOG2 gene and CMVD and suggest a potential mechanism whereby FOG2 may regulate HIF target gene expression. Future studies are needed to further characterize potential interactions between FOG2 and

HIFs and to understand the role of altered angiogenic gene expression on coronary microvascular structure and function.

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Disclosures

None.

Supplemental Material

Supplemental Methods
Tables S1–S3
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